Multimodal Misinformation Detector

Three tracks, one question: what does each kind of evidence really tell us?

Harika Etikela
Text track

Ananya
Image track

Rafiul Haider
Combined track

CS 668 Analytics Capstone · Pace University · Midterm

One problem, three ways it spreads



False text

The claim in the article is wrong

Text track
Harika Etikela

WELFake



Edited or generated image

The picture itself is fake

Image track
Ananya

CIFAKE



Real image, wrong caption

Both are real; the pairing misleads

Combined track
Rafiul Haider

MMFakeBench · VERITE · NewsCLIPpings

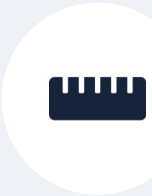
Each track has its own dataset and task, so scores are read within a track, never across tracks.

How we work, in every track



Clean the data first

Remove duplicates and leaks before any test



Start from a floor

Know what simple guessing scores



Measure uncertainty

95% ranges, not single numbers



Say what it can't do

Limits travel with every result

Real and fake articles leave different clues

Text track · Harika Etikela

63,678

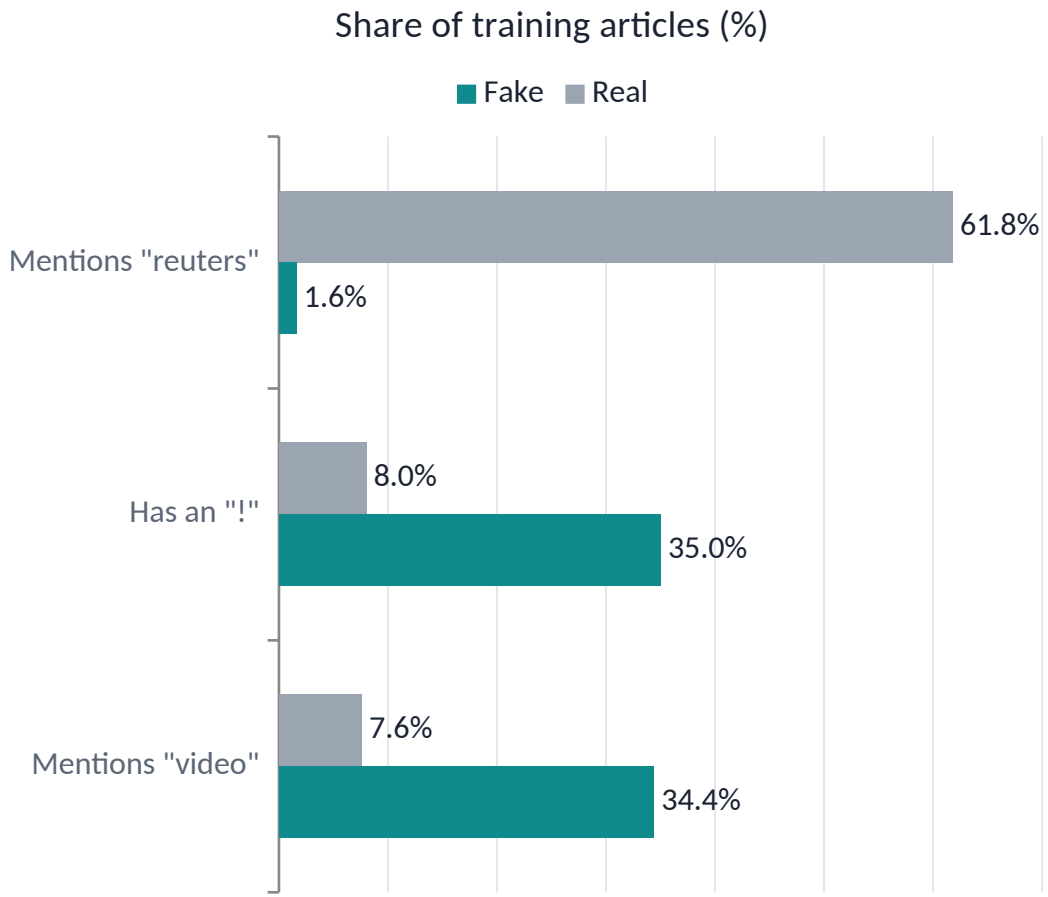
articles after removing 8,456 duplicates

70 · 15 · 15

train · validation · test split

54.6%

accuracy from always guessing "Real"



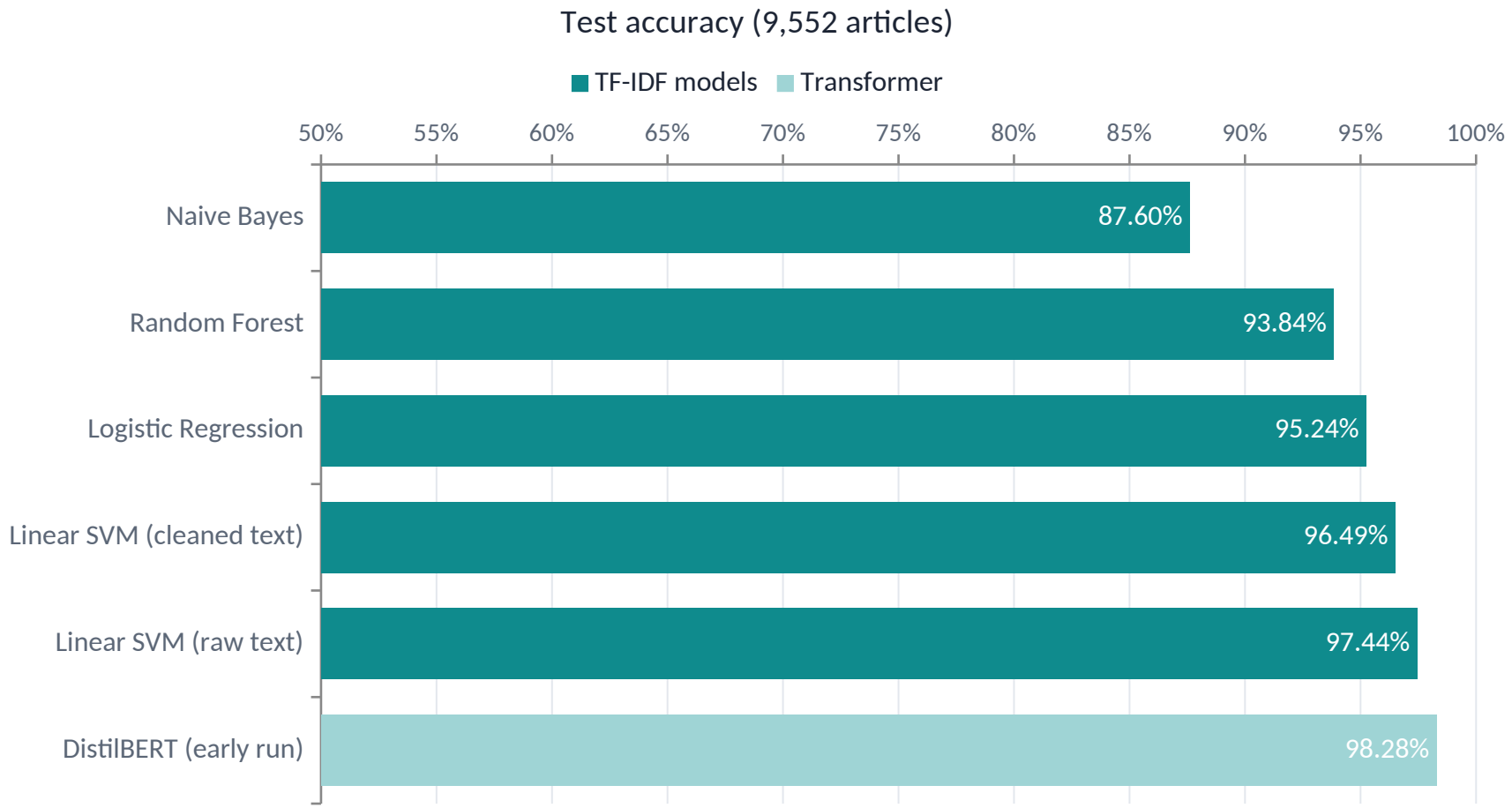
Good to know

Strong source and style cues mean a model can learn the outlet's style, not only the facts.

Text track midterm results, Harika Etikela (WELFake training split).

From simple to deep models

Text track · Harika Etikela



Good to know

DistilBERT is an early run: 5,000 training articles, one epoch. It does not yet show transformers are better.

Text track midterm results, Harika Etikela. Axis starts at 50%.

Words carry the signal, partly through style

Text track · Harika Etikela

97.4% → 96.2%

Linear SVM test accuracy with the word "reuters" removed

About 1.3 points come from a single source word.

Next for the text track

1. **Ensemble**
   SVM + DistilBERT, soft voting
2. **Explanations**
   LIME and SHAP feature comparison
3. **New datasets**
   ISOT and LIAR, no retraining

Good to know

Results so far report accuracy; precision, recall and F1 come with the next phase.

Text track midterm results, Harika Etikela.

Clean first, then test

Image track · Ananya

120,000

CIFAKE images, 32×32
half real, half generated

Duplicate audit

2,218 train · 454 test
378 shared across splits

Clean split

10,000 · 1,996 · 3,986
0 images shared with train

Train · validation · test images in the clean working subset

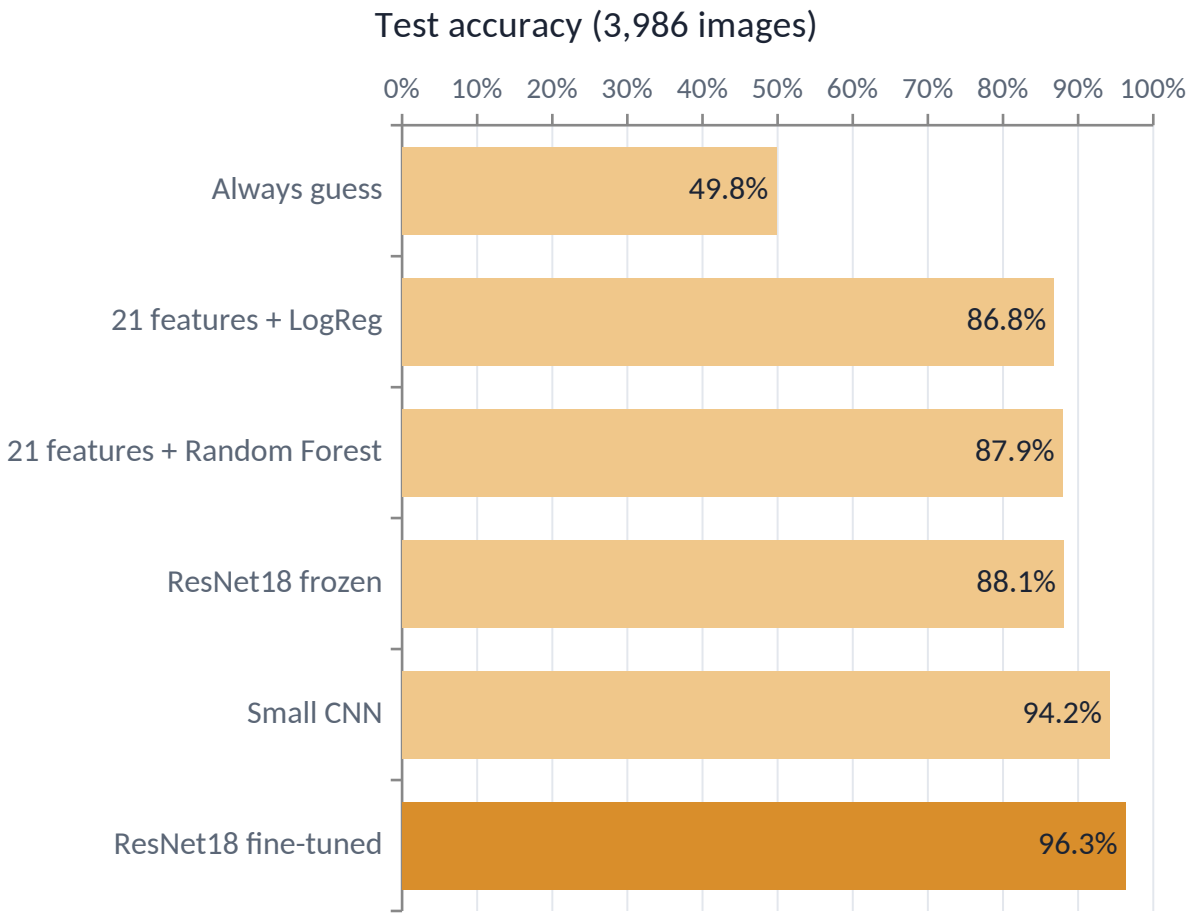
Good to know

Pixel-identical copies were found across splits and removed before the final run, so test scores are not inflated.

Image track midterm results, Ananya (CIFAKE: Bird and Lotfi, 2024).

Climbing the model ladder

Image track · Ananya



96.3%

fine-tuned ResNet18 accuracy

0.995

AUC for the same model

39×

smaller small CNN, only 2.1 points behind

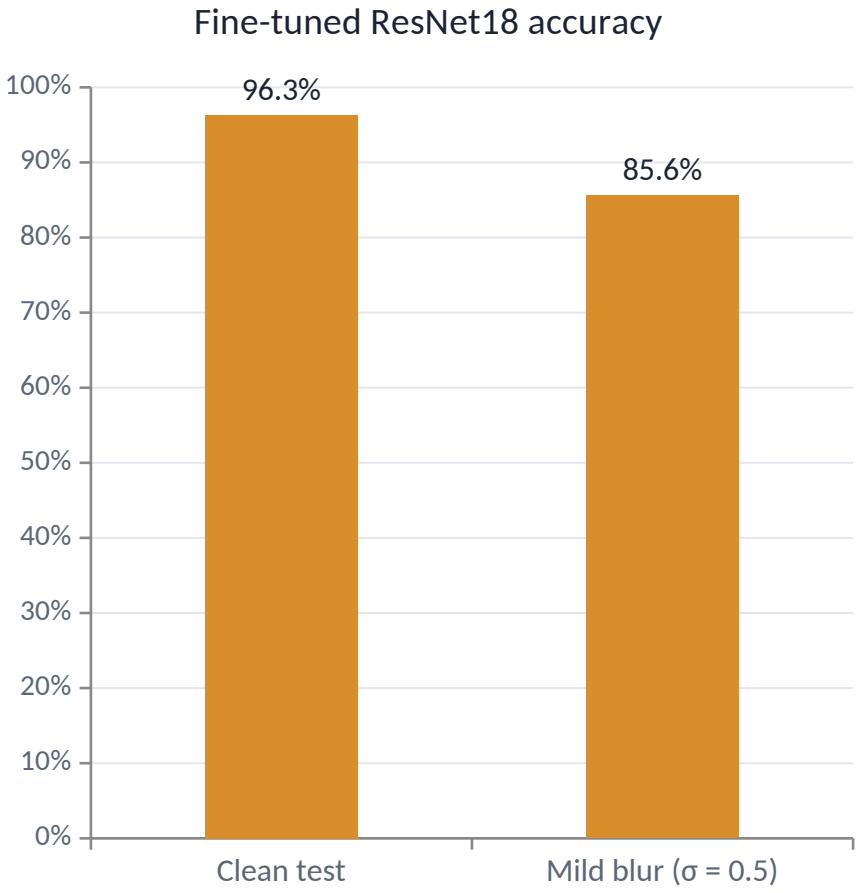
Good to know

One run so far; three seeds and the full training set are planned.

Image track midterm results, Ananya. 95% range for ResNet18: 95.7% to 96.8%.

Fine detail, not color, carries the signal

Image track · Ananya



Strong blur (σ = 1)
AUC 0.995 → 0.77

Grayscale
AUC stays 0.98

JPEG quality 75
No change

Next: test on images from other generators and real-photo sources.

Good to know

Grad-CAM shows where the model looks; it is not proof of a true generator fingerprint.

Image track midterm results, Ananya.

What does a pretrained model already know?

Combined track · Rafiul Haider

Image + caption → LAYA-Vision → Answer

Frozen: no training, no prompt tuning

Good to know

This is the reference point a fine-tuned model must beat later.

MMFakeBench
10,000 test rows
4 labels

VERITE
1,001 rows
3 labels

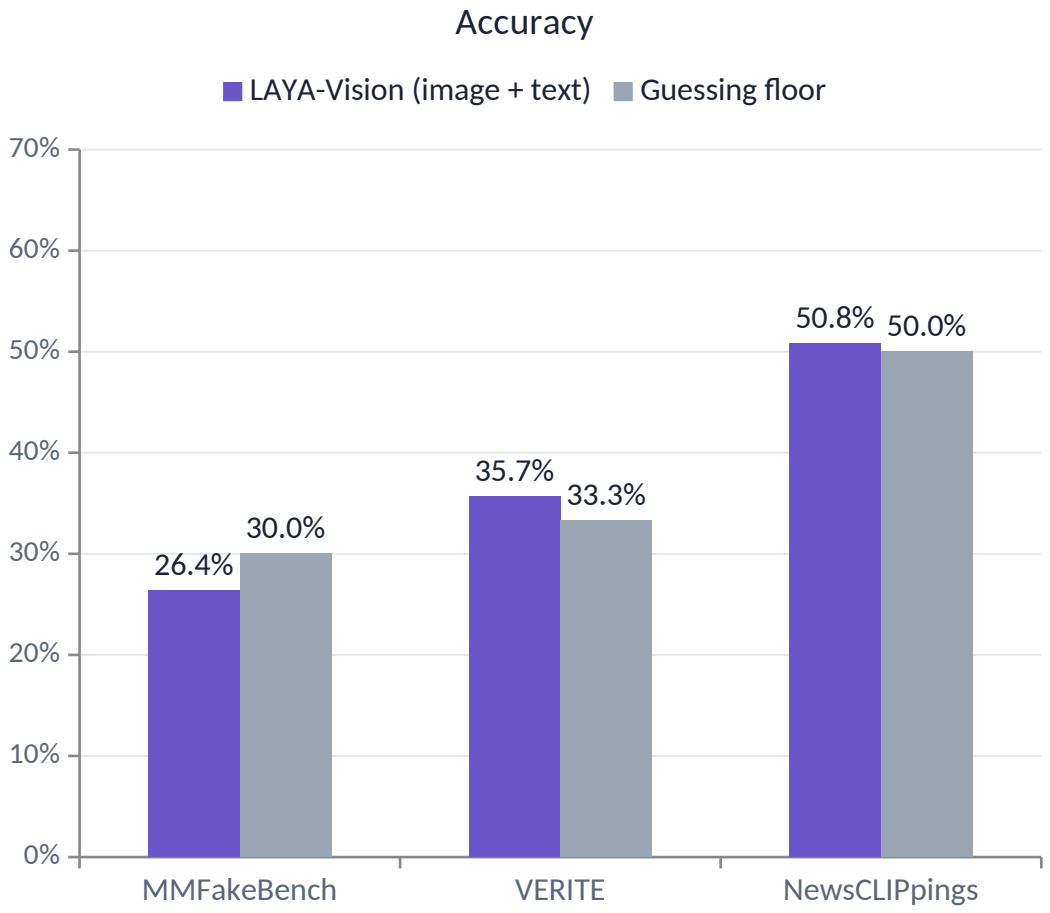
NewsCLIPpings
7,024 pairs
2 labels

Same rows, three ways: image + text · text only · image only

Combined track, Rafiul Haider. LAYA-Vision checkpoint f2fe3c12, one fixed question set.

Asked cold, it mostly guesses

Combined track · Rafiul Haider



91.5%

of MMFakeBench labels guessed from source names alone, without seeing the image or text

−1.5 pts

average change when the model also sees the image (from −18 to +18 by source)

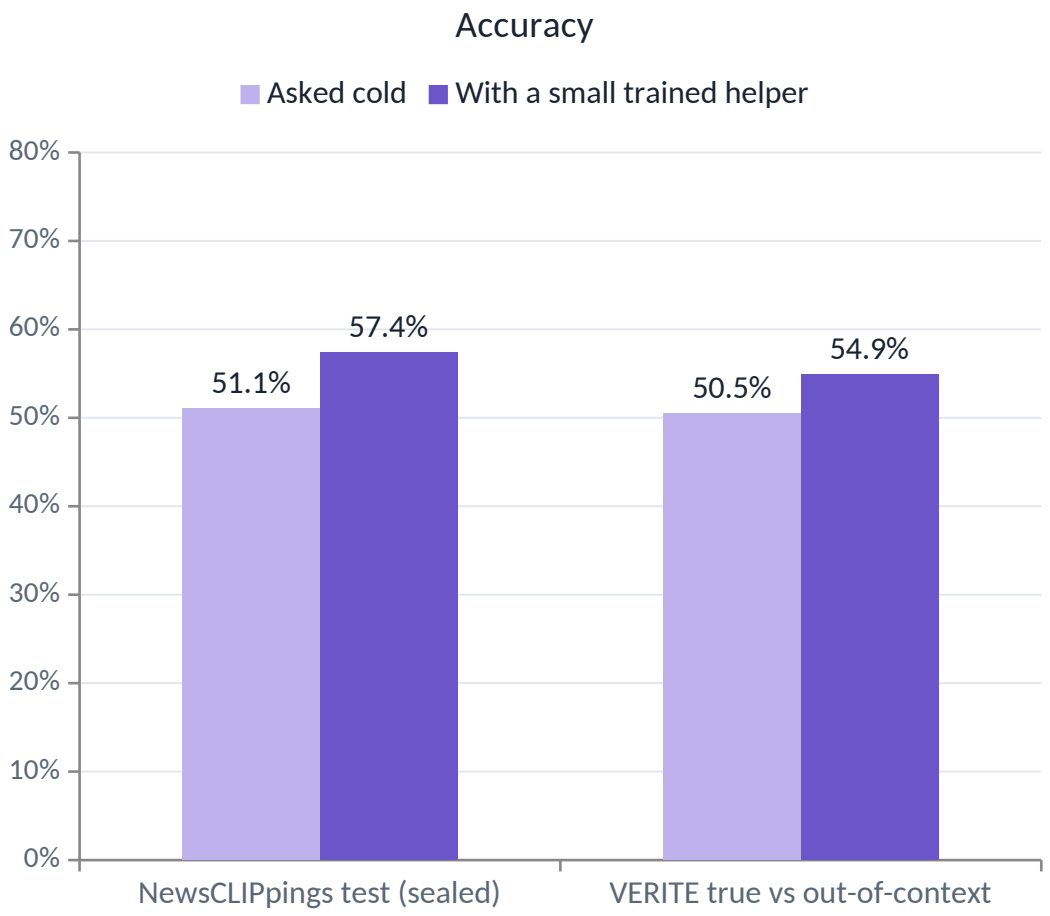
Good to know

The MMFakeBench test split had been looked at before, so its numbers are a check-up, not a final exam.

Combined track, Rafiul Haider. n = 9,999 · 1,001 · 7,024. Floors: always-majority, 1 in 3, coin flip.

The signal is there, just unused

Combined track · Rafiul Haider



+6.2 pts

on 7,264 sealed test pairs, read once (95% range +4.8 to +7.7)

The helper learns from 71,072 NewsCLIPpings examples on top of the frozen model; the model itself never changes.

Good to know

On VERITE the gain (+4.4 pts) sits inside its 95% range of zero, so it may be luck.

Combined track, Rafiul Haider. Helper = linear probe on frozen 576-d LAYA-Vision features.

What the three tracks teach together



Text track

Strong signal from words

Part of it is outlet style ("reuters")



Image track

Strong on its own benchmark

Mild blur costs about 11 points



Combined track

The signal sits inside the model

Source names give MMFakeBench labels away

Every benchmark hides a shortcut. Finding it is part of the result.

Road to the final

Text track	Ensemble	LIME / SHAP	ISOT and LIAR
Image track	Unseen generators	Three seeds	Full training set
Combined track	Attention and Gate heads	CLIP classifier	Three seeds

Good to know

The combined track's stretch targets (84.8% and 76.3%) come from fully trained systems and are not met yet.

One shared output format → one app that gives a verdict for text, image or both

Thank you

Questions welcome



Harika Etikela
Text track



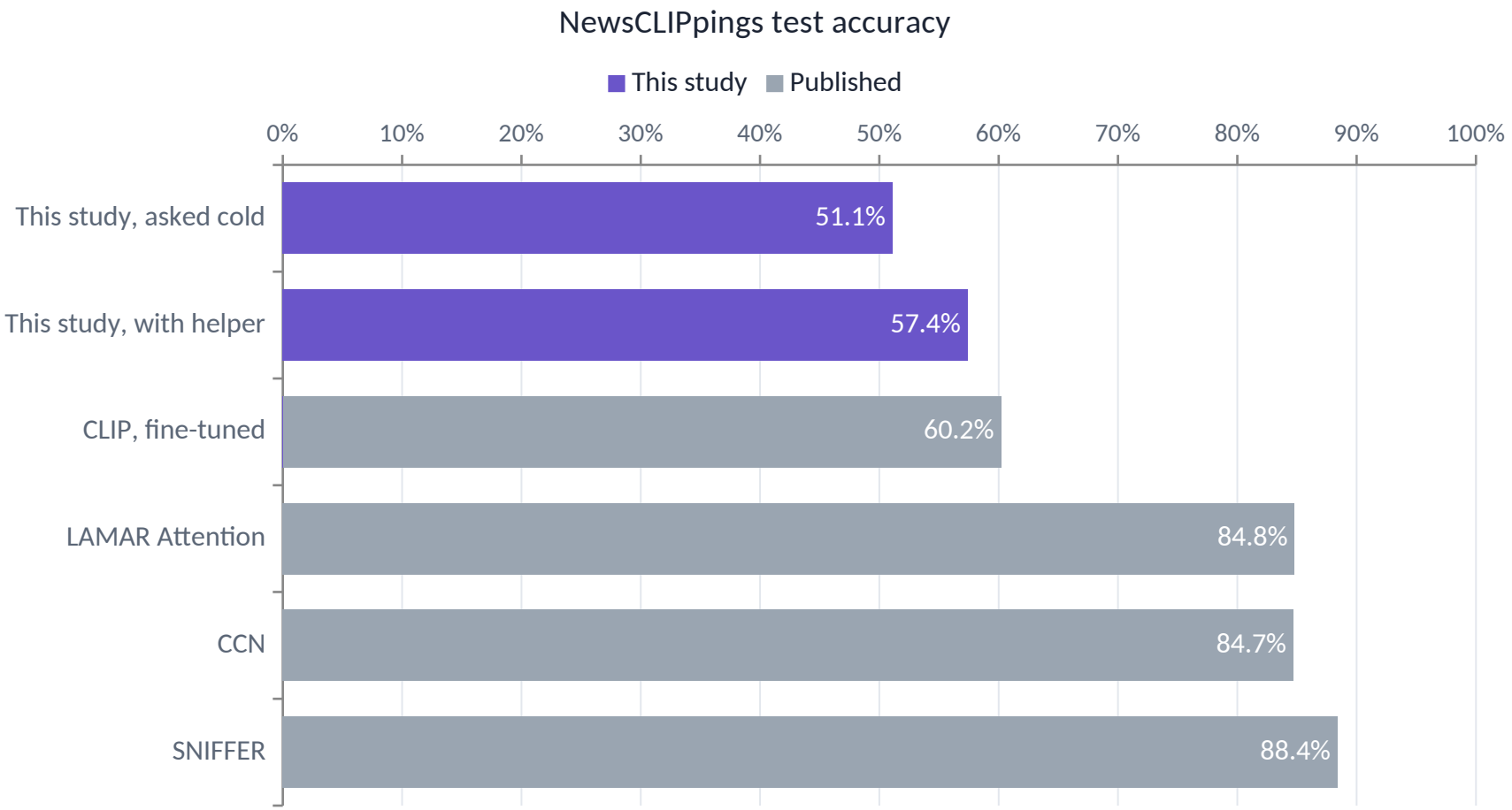
Ananya
Image track



Rafiul Haider
Combined track

Backup: published systems on NewsCLIPpings

Combined track · Rafiul Haider



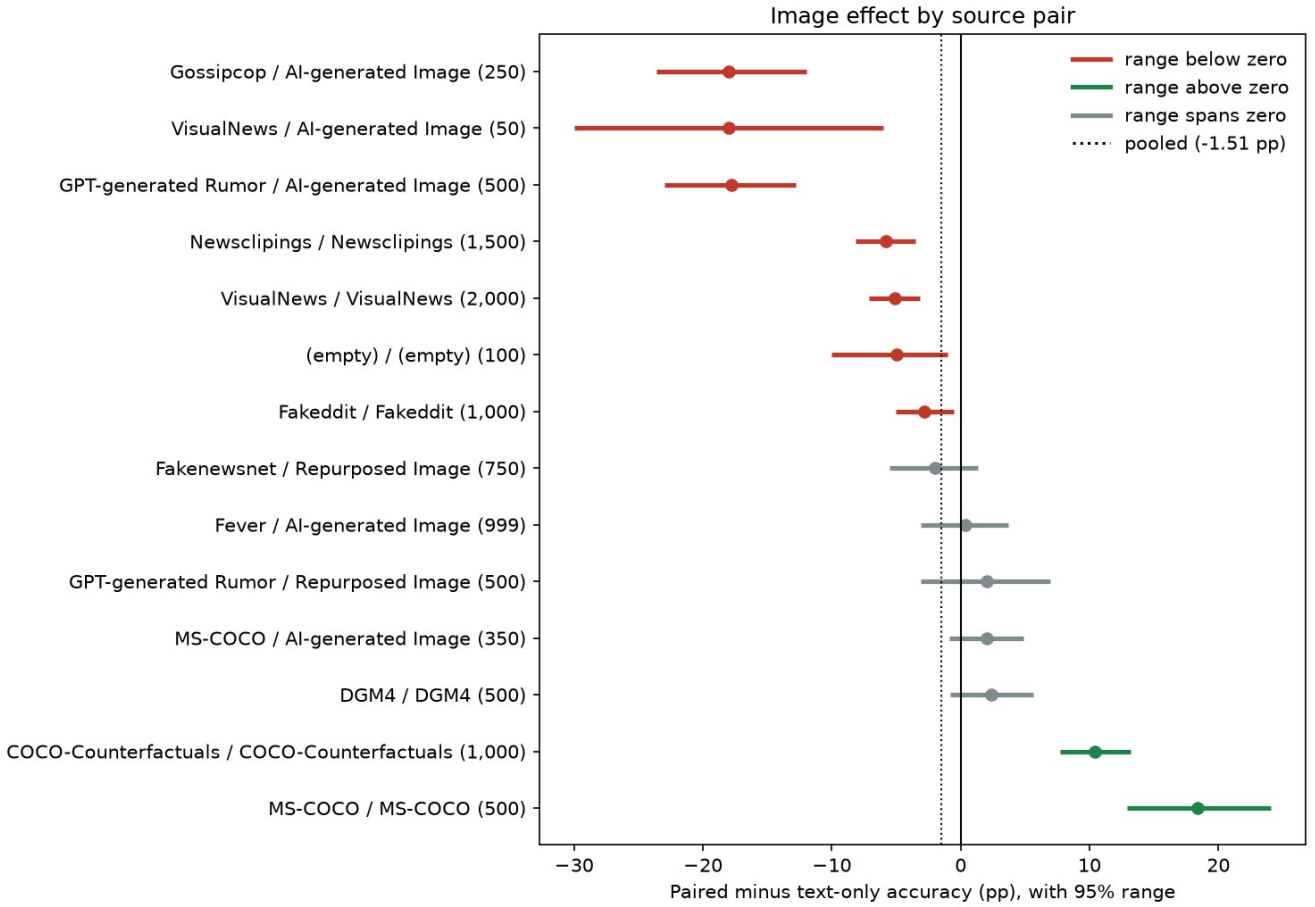
Good to know

Published systems retrain the whole model; CCN and SNIFFER also search the web. These are reference points, not a ranking.

Values checked against each paper's results table on 25 Oct 2026.

Backup: the image effect by source

Combined track · Rafiul Haider



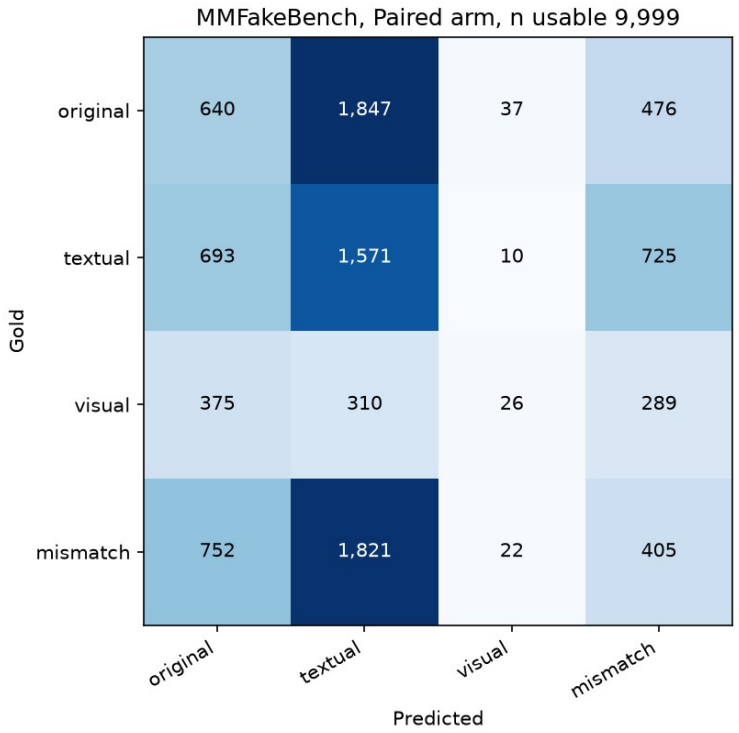
Good to know

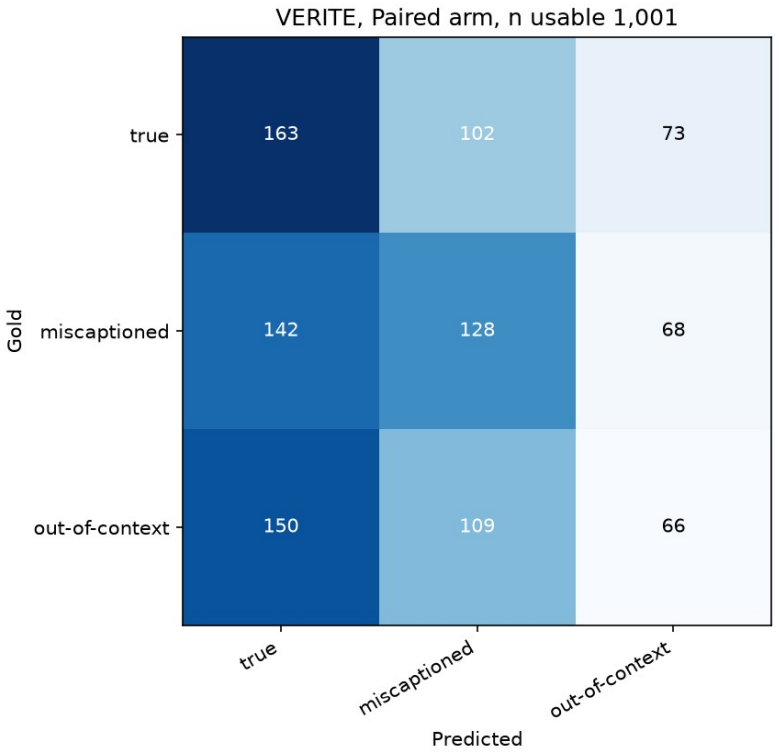
On 13 of 14 sources every row has the same label, so the image mainly changes which answer the model picks.

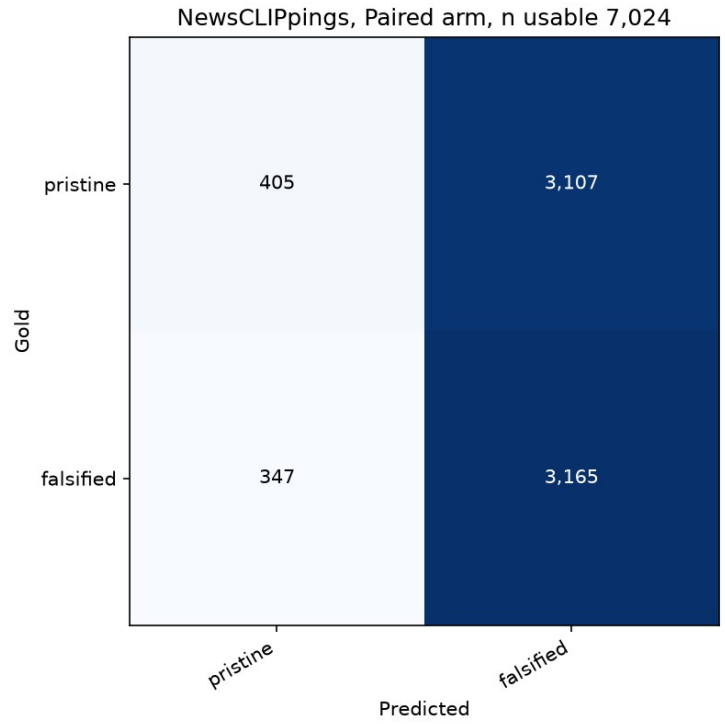
MMFakeBench test, 9,999 rows, paired minus text-only accuracy with 95% ranges.

Backup: where the model's answers go

Combined track · Rafiul Haider







Zero-shot, image + text. MMFakeBench test (9,999), VERITE (1,001), NewsCLIPpings validation (7,024).

Backup: where to find everything

Showcase folder
MIDTERM_PRETRAINED_BASELINE/README.md

Executed notebooks
notebooks/01 to 04, each with self-checks

Detailed dashboard
http://localhost:8501

Plain-language dashboard
http://localhost:8502